

Raghav Nandwani

+91-9999003348 | raghavnandwani@hotmail.com | linkedin.com/in/raghavnandwani
github.com/rvnandwani | rvnandwani.github.io

Engineering Manager | Physical AI | Applied AI | Robotics Software | Multimodal AI

Professional Summary

Engineering Manager and Lead AI & Robotics Engineer with 6+ years of experience building production AI systems spanning robotics, multimodal AI, speech technologies, computer vision, and edge intelligence. Proven experience leading cross-functional teams delivering Physical AI platforms, autonomous robotics software, AI infrastructure, and GPU-accelerated inference across cloud and embedded environments. Expertise includes ROS 2, C++, Python, CUDA, TensorRT, NVIDIA Isaac ecosystem, transformer models, computer vision, LLM-powered applications, and MLOps. Passionate about translating research into scalable production systems deployed across intelligent devices, autonomous robots, and multimodal AI platforms.

Core Technical Expertise

- Programming: Python, C++17, SQL, Linux
- Applied AI: PyTorch, TensorFlow, Transformers, LLMs, Speech AI, Computer Vision, Multimodal AI
- Robotics: ROS 2, Isaac ROS, Visual SLAM, Localization, Navigation, Sensor Fusion, NVBlox, Motion Planning, GR00T
- GPU & Edge AI: CUDA, TensorRT, NVIDIA Jetson AGX Orin, Xavier, Quantization, Performance Optimization
- Platforms: Docker, Kubernetes, Databricks, Git, CI/CD, MLOps

Professional Experience

Jio Platforms (Previously Tesseract)	Oct 2024 – Present
Engineering Manager – Core AI	Mumbai, India

- Lead the Core AI organization within Jio's Physical AI & Humanoid Robotics division, driving architecture and delivery across robotics software, multimodal AI, speech systems, and embodied intelligence.
- Manage multidisciplinary teams spanning ROS 2, Applied AI, Computer Vision, Platform Engineering, and Embedded AI while defining technical strategy and execution.
- Architect production software platforms spanning ROS 2 robotics, perception, localization, navigation, and low-latency Voice-to-Voice AI systems deployed across intelligent devices.
- Evaluated, fine-tuned, benchmarked, and deployed AI models including Wake Word Detection, LID, STT, TTS, Voice Cloning, VQA, and LLM-powered orchestration.
- Improved Indian language identification accuracy from 78% to 92% through model development and benchmarking.
- Built a scalable Voice-to-Voice AI platform integrating ASR, Speaker Diarization, LLMs, and TTS supporting over 10,000 requests/hour with architecture designed for million-scale daily traffic.
- Designed Visual SLAM, localization, occupancy mapping, and 3D reconstruction pipelines using Isaac ROS, CUDA, TensorRT, and NVBlox, improving localization throughput by 20%.
- Established Databricks-based MLOps, observability, analytics, semantic caching, personalization, and AI lifecycle infrastructure while mentoring engineers and driving engineering excellence.

Qualcomm Technologies	Mar 2023 – Dec 2023
Senior Engineer – XR Research	San Diego, USA

- Conducted applied research in multimodal perception, scene understanding, and edge AI under real-time latency and power constraints.
- Improved detection and segmentation accuracy by approximately 25% using transformer-based architectures and multimodal feature fusion.
- Benchmarked AI models across latency, accuracy, and hardware efficiency for deployment on edge accelerators.
- Collaborated with AI researchers, hardware architects, and systems engineers while mentoring engineers on experimentation and optimization.

Monarch Tractor
Autonomy Engineer Intern

Nov 2022 – Mar 2023
California, USA

- Developed autonomous perception, localization, and navigation software for agricultural robots.
- Optimized robotics software for NVIDIA Jetson Xavier using latency profiling and hardware-aware optimization.
- Validated perception and autonomy performance through real-world field testing in challenging outdoor environments.

Gift A Life
Senior Data Scientist

Feb 2024 – Oct 2024
Delhi, India

- Deployed production computer vision models and improved inference efficiency through profiling, optimization, and MLOps practices.

Uniphore
Data Scientist

Aug 2020 – Aug 2022
Dallas, USA

- Developed production computer vision systems emphasizing scalability, robustness, and low-latency inference.
- Collaborated with product and engineering teams to deliver customer-facing AI solutions.

Education

University of Maryland, College Park
Master of Engineering, Robotics & Artificial Intelligence

2018–2020

Manipal University Jaipur
Bachelor of Technology, Mechatronics Engineering

2013–2017

Certifications

- Udacity Sensor Fusion Nanodegree
- AWS Technical Essentials
- AWS Machine Learning Pipelines on Cloud